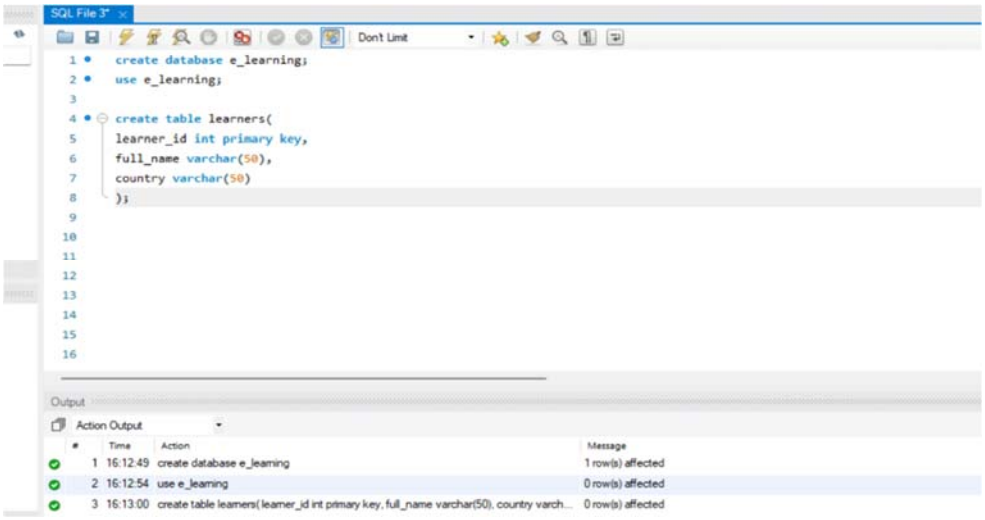
Module End Assignment

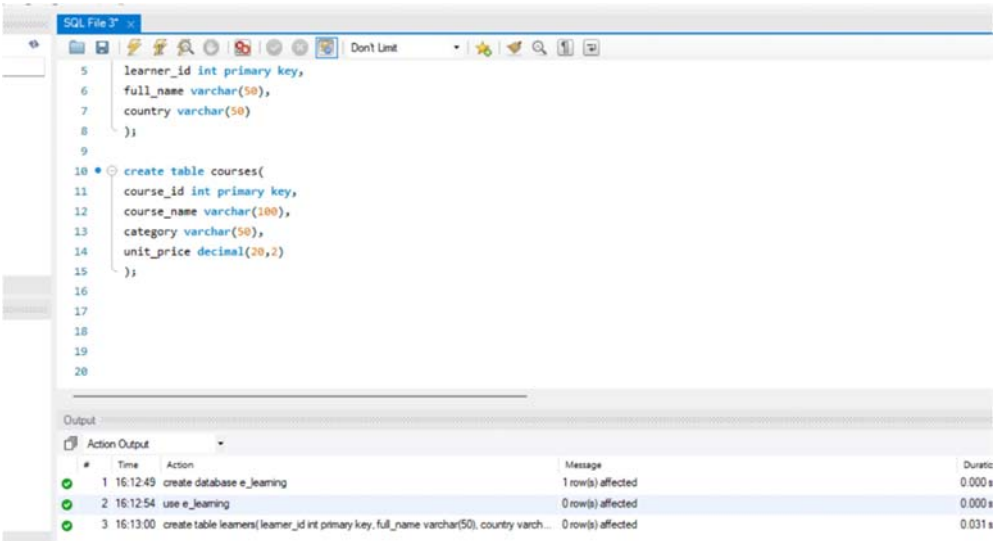
Analyzing E-Learning Platform Purchases using MySQL

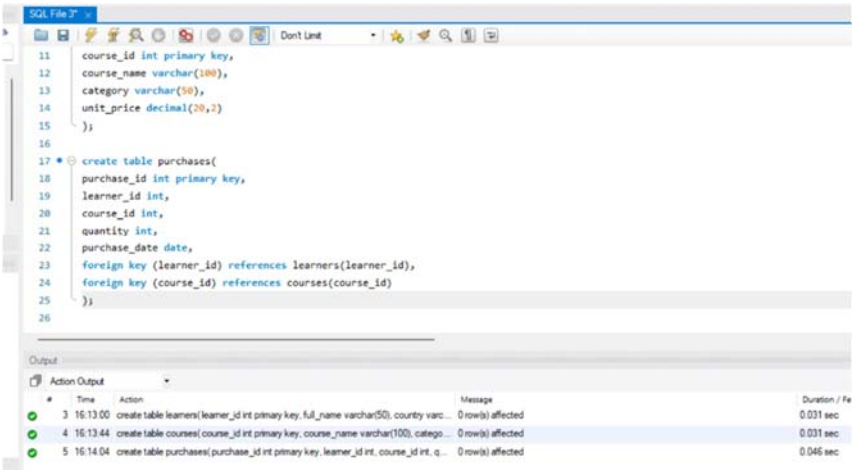
Database Creation and Schema Design

Created a MySQL database named **e_learning** for the E-Learning Course Purchase Analysis project. Designed and created three tables**: Learners, Courses** and **Purchases**, using appropriate data types and constraints.

Established primary key and foreign key relationships between the tables to maintain data integrity and connect learner, course, and purchase information.

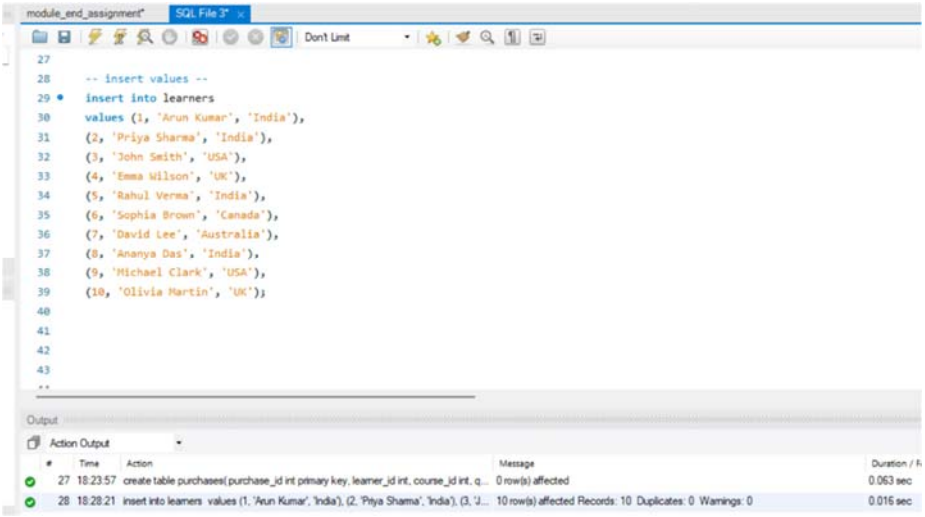


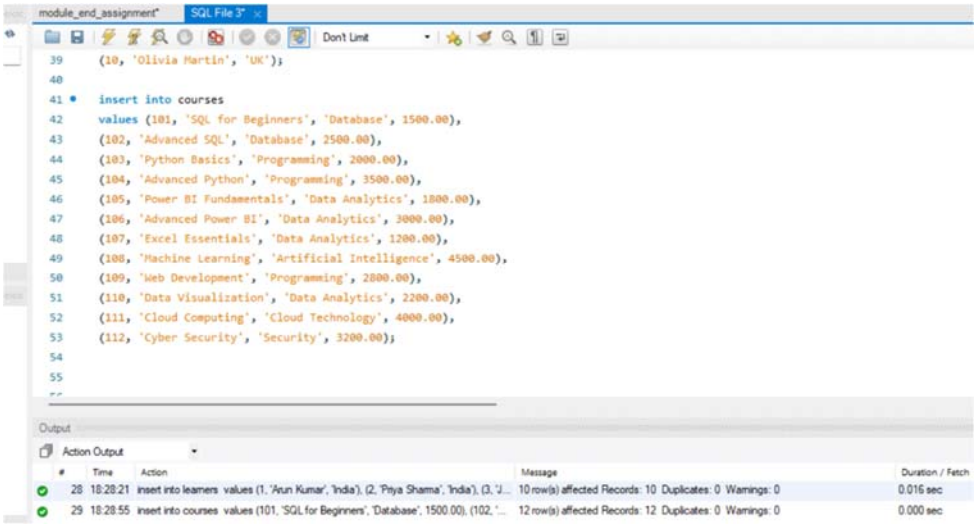


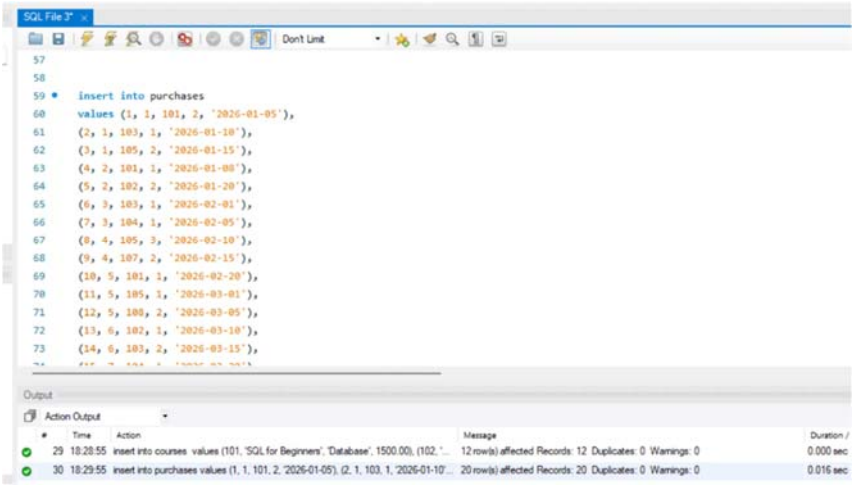


Insert sample data

Sample records were inserted into all three tables (Learners, Courses, and Purchases) using SQL INSERT statements.

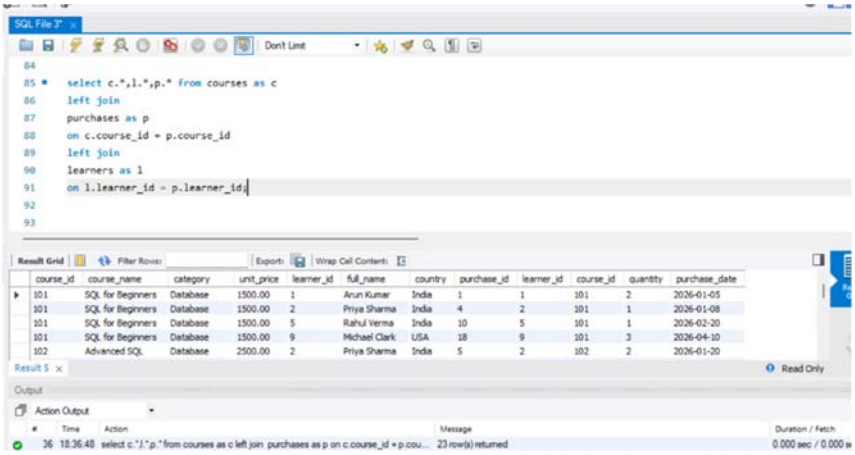




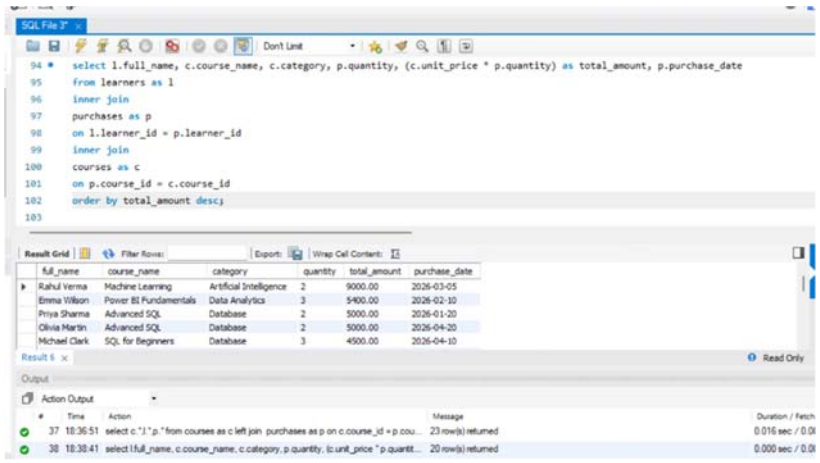


Data Exploration Using Joins

Combined learner, course, and purchase data using a **LEFT JOIN** to retrieve learner details along with their purchased course information, including course name, category, quantity, and purchase date.

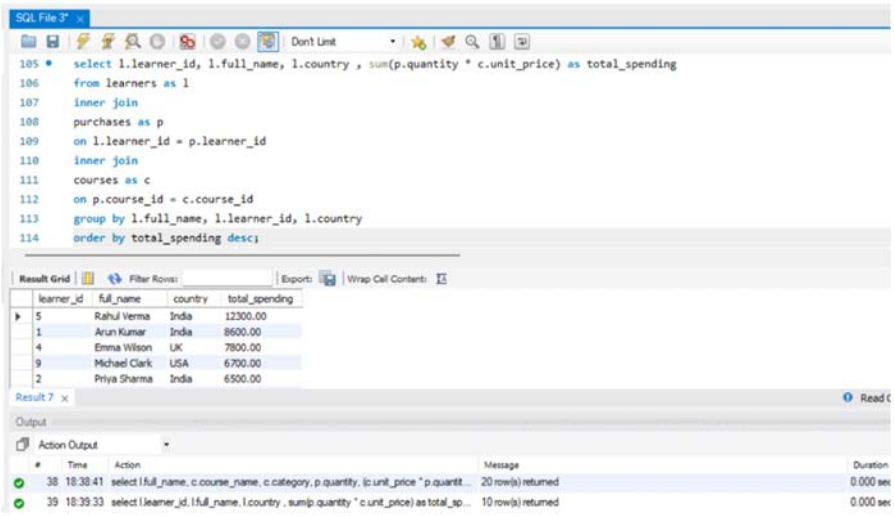


Retrieved learner-wise purchase details by combining the three tables using **INNER JOIN**, including course information, quantity, total amount, and purchase date.

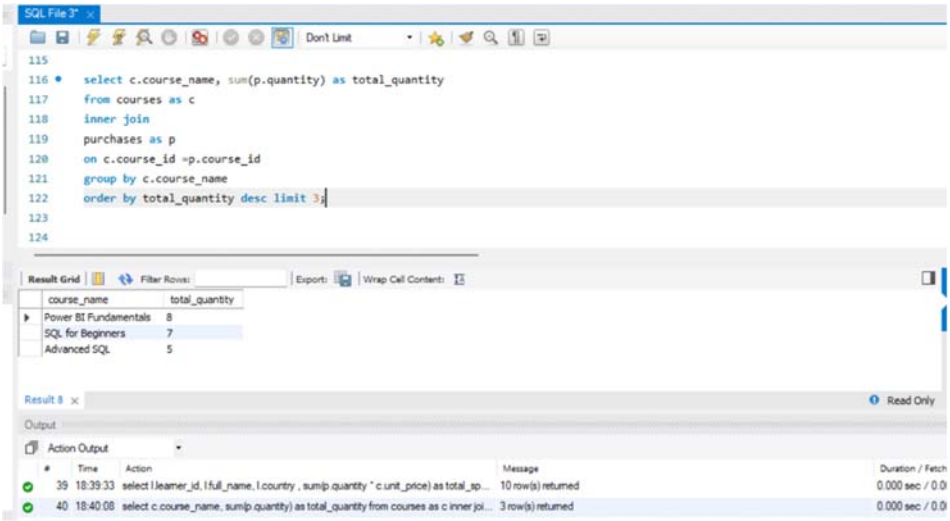


Analytical Queries

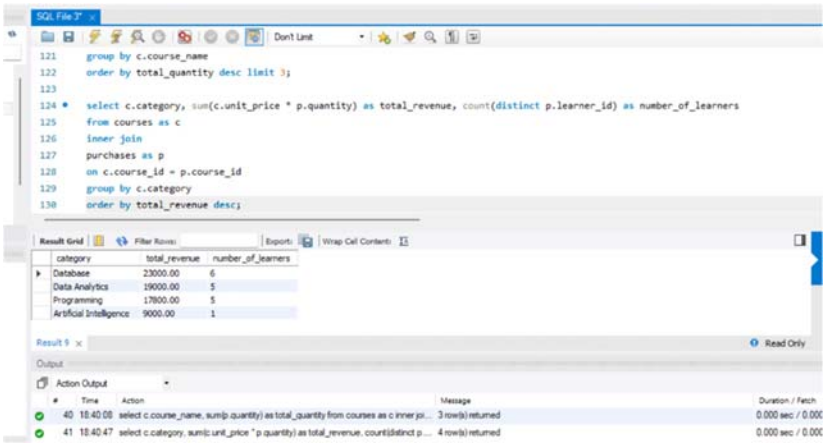
Calculated the total spending of each learner by multiplying the course purchase quantity by the unit price and aggregating the results using the **SUM()** function. Learner details, including their names and countries, were retrieved by joining the Learners, Purchases, and Courses tables using **INNER JOIN**. The results were grouped by learner and sorted in descending order of total spending to identify the highest-spending learners.



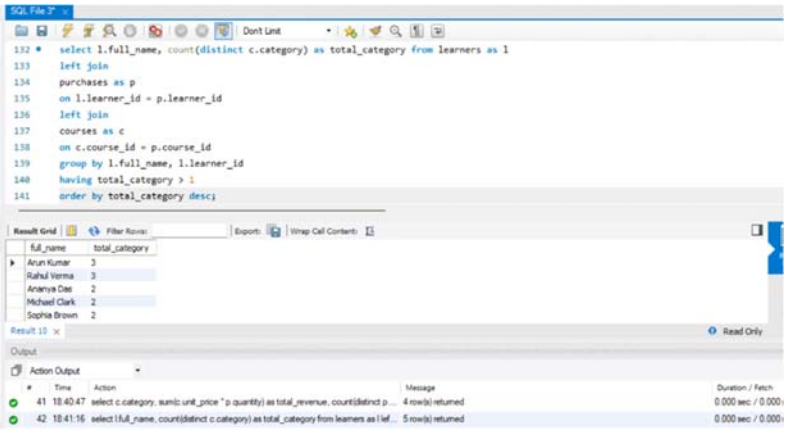
Identified the top 3 most purchased courses by calculating the total quantity sold for each course using **INNER JOIN, SUM(), GROUP BY** and **ORDER BY** clauses. The results were sorted in descending order to display the three courses with the highest purchase quantities.



Calculated the total revenue and number of unique learners for each course category using INNER JOIN, aggregate functions, and GROUP BY, and displayed the results in descending order of revenue.



Identified learners who purchased courses from more than one category by joining the Learners, Purchases, and Courses tables using **LEFT JOIN.** Used **COUNT(DISTINCT)** to calculate the number of unique course categories for each learner and applied the HAVING clause to filter learners with more than one category. The results were sorted in descending order based on the total number of categories.



Identified courses that have not been purchased by using a **LEFT JOIN** between the Courses and Purchases tables and filtering records with **NULL** values in the Purchases table.

